

SEAL: Fault-Tolerant Neuron Model for Spiking Neural Networks on Unreliable DRAM Systems

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Presentation Outline

- Introduction
- Proposed Methodology
- Evaluation
- Conclusion and Future Work
- References



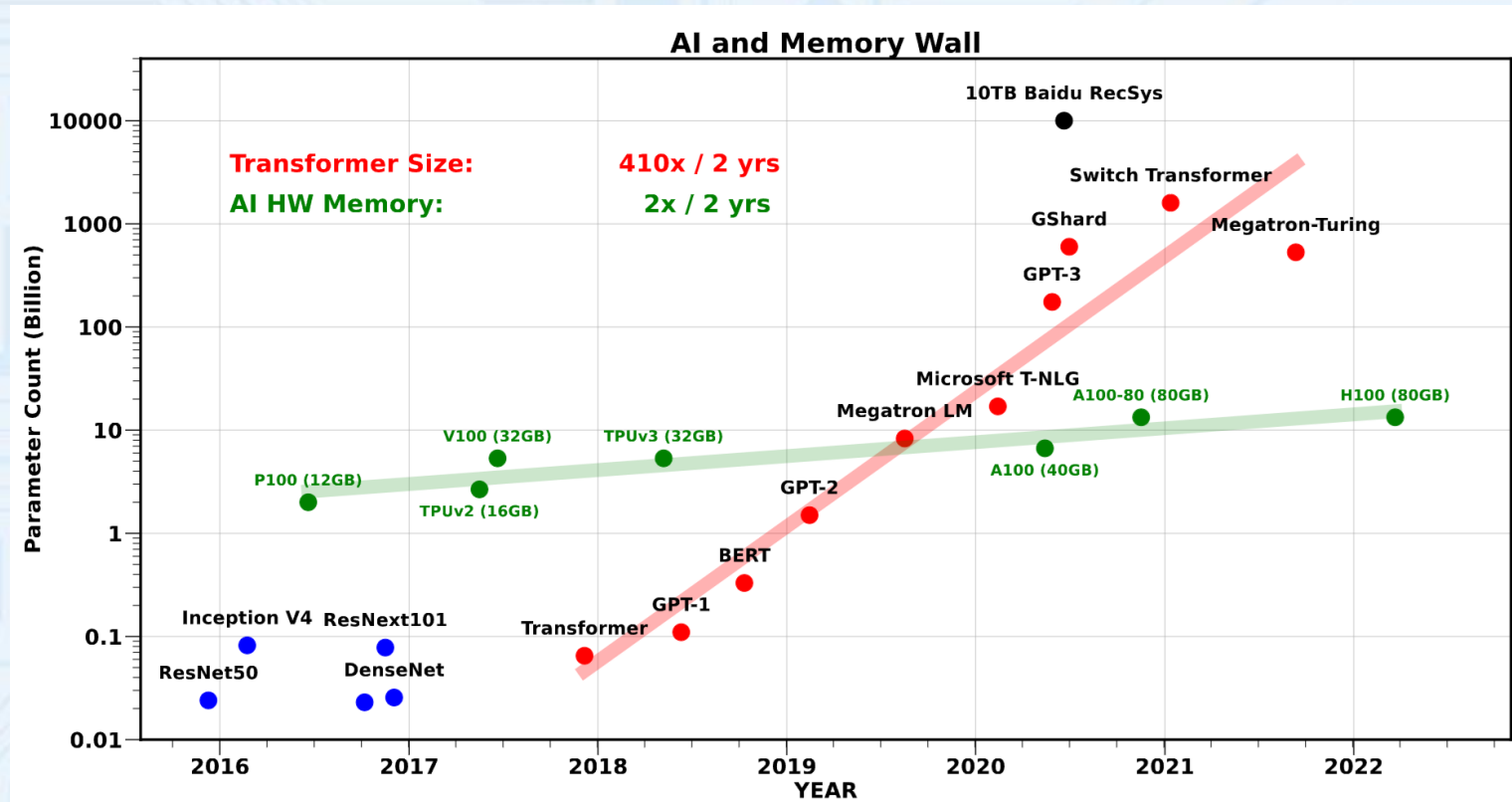
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Introduction

- AI workloads demand intense memory access
 - E.g., GPT-3 has over 100 billion parameters



A. Gholami et al., Micro'24 [1]



Introduction

- 3D-DRAM provides high-bandwidth & low-latency memory access
 - E.g., HBM3 provides up to 0.7TB/s

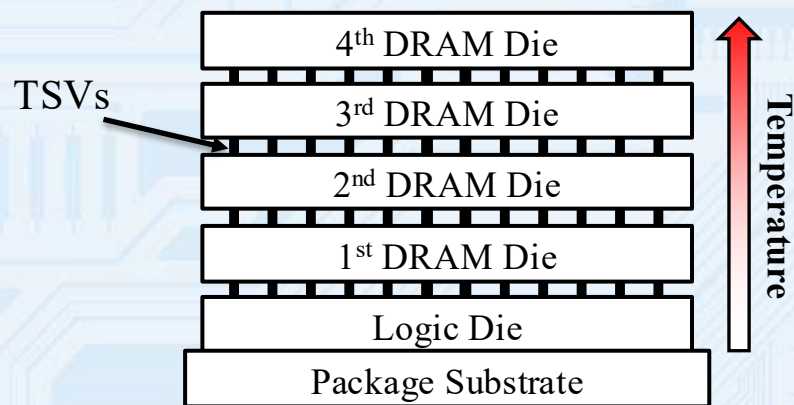
Category	Item	HBM1	HBM2	HBM2E	HBM3	HBM3E	HBM4
	Time	2014	2018	2020	2022	2024	2026
General	Die Density	2Gb	8Gb	16Gb	16Gb	24Gb	24Gb
	Max. Bandwidth	128GB/s	0.3TB/s	0.5TB/s	0.7TB/s	1.18TB/s	1.65TB/s
Configuration	Max. Stack Height	4HI	8HI	8HI	12HI	12HI	16HI
	Capacity	1GB	8GB	16GB	24GB	36GB	48GB
	Total IO/Cube	1024	1024	1024	1024	1024	2048
Power	vDDC	1.2V	1.2V	1.2V	1.1V	1.1V	1.05V
	vPPE	2.5V	2.5V	2.5V	1.8V	1.8V	1.8V
	vDDQ	1.2V	1.2V	1.2V	1.1V	1.1VV	0.8V

K. Kim and M. Park, IMW'24 [2]

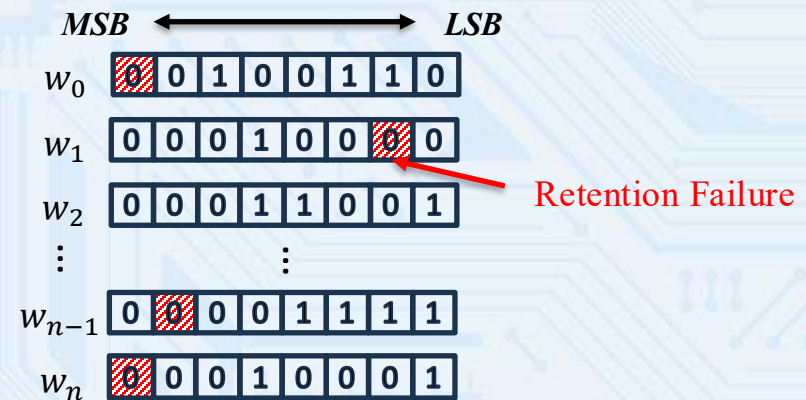


Motivation & Contribution

- High accumulation exacerbates retention failure
 - Data collapse in network parameters
 - E.g., Bit-flipping in weights



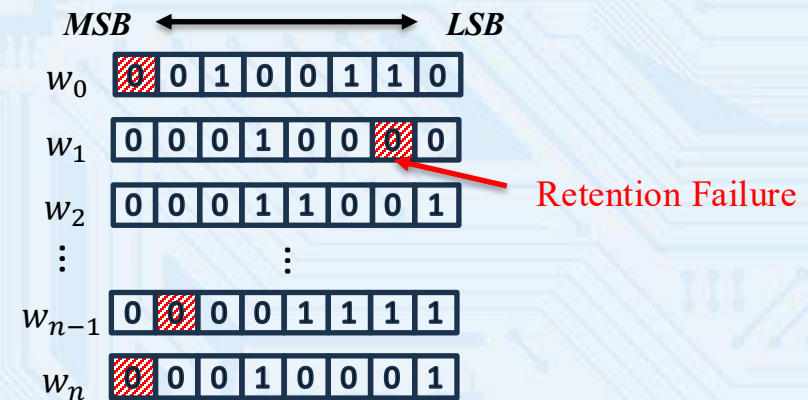
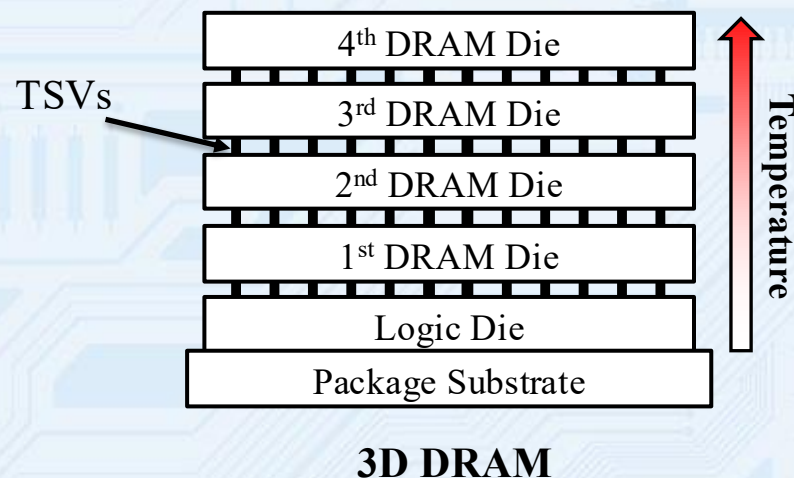
3D DRAM



Example of Bit-Flipping

Motivation & Contribution

- High accumulation exacerbates retention failure
 - Data collapse in network parameters
 - E.g., Bit-flipping in weights
- **Self-Error-Adaptive LIF (SEAL)**
 - Neutralize retention failure at runtime
 - Minimize modification from LIF neuron



Example of Bit-Flipping

Why SEAL?

- Prior works mostly rely on *synapse*-based fault recovery [3-5]
 - Exponential scaling of HW overhead
- SEAL is a *neuron*-based approach
 - Linear scaling of HW overhead
- Hardware-friendly & network agnostic fault-tolerance mechanism
 - With the best error recovery among SOTAs



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Proposed Methodology

- Semantics of retention failure
 - Fractional $\Rightarrow$ **Decrease**

Fractional

$0101\underline{1}001 \xrightarrow{1 \rightarrow 0} 0101\underline{0}001$
Sign Fractional

$10110\underline{1}00 \xrightarrow{1 \rightarrow 0} 10110\underline{0}01$
Sign Fractional



Proposed Methodology

- Semantics of retention failure
 - Fractional => **Decrease**
 - Sign-bit => **Increase**

Fractional

$0101\underline{1}001 \xrightarrow{1 \rightarrow 0} 0101\underline{0}001$
Sign Fractional

$10110\underline{1}00 \xrightarrow{1 \rightarrow 0} 10110\underline{0}01$
Sign Fractional

Sign-bit

$\underline{1}0110100 \xrightarrow{1 \rightarrow 0} \underline{0}0110001$
Sign Fractional



Proposed Methodology

- Store with **inverted sign-bit**
 - Fractional => **Decrease**
 - Sign-bit => **Decrease**

Fractional

$0101\underline{1}001 \xrightarrow{1 \rightarrow 0} 0101\underline{0}001$
Sign Fractional

$10110\underline{1}00 \xrightarrow{1 \rightarrow 0} 10110\underline{0}01$
Sign Fractional

Sign-bit

$\underline{0}0110100 \longrightarrow \textcircled{1}0110100 \xrightarrow{1 \rightarrow 0} \underline{0}0110001 \longrightarrow \textcircled{1}0110001$
Sign Fractional **Inverted** **Inverted**



Proposed Methodology

- SEAL is designed to stimulate network activity
 - $S_j(t)$: Sum of incoming spikes at t
 - D_j : Average weight distortion to neuron j

$$u_j(t+1) = \tau_m^{-1}u(t) + \underbrace{\sum_i^n w_{i,j}s_i(t) - s_j(t)V_{th}^j}_{\text{LIF equation}} + \varepsilon_j$$

Proposed error adaptive term
↓

where, $\varepsilon_j = S_j(t)D_j$ such that

$$S_j(t) = \sum_i^n s_i(t), \quad D_j = \frac{1}{n} \left(\sum_i^n \underbrace{w_{i,j}}_{\text{Correct weight}} - \sum_i^n \underbrace{\bar{w}_{i,j}}_{\text{Affected weight}} \right)$$

Affected weight

Proposed Methodology

- Virtual Synaptic Weight (VSW) to minimize the modification from the LIF neuron

$$u_j(t + 1) = \tau_m^{-1} u(t) + \underbrace{\sum_i^n w_{i,j} s_i(t) - s_j(t) V_{th}^j}_{\text{LIF equation}} + \varepsilon_j$$

Proposed error adaptive term
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Proposed Methodology

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Proposed error adaptive term
↓

⇓ Introducing VSW

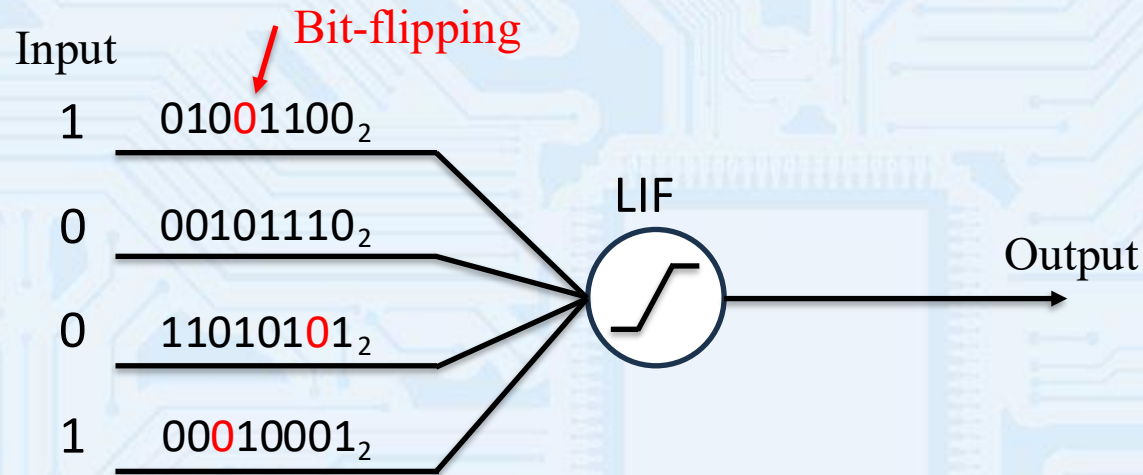
$$u_j(t+1) = \tau_m^{-1}u(t) + \underbrace{\sum_i^{n+1} \bar{w}_{i,j}s_i(t) - s_j(t)V_{th}^j}_{\text{LIF equation}}$$

where, $\bar{w}_{i,j} = \varepsilon_j = s_j(t)D_j$



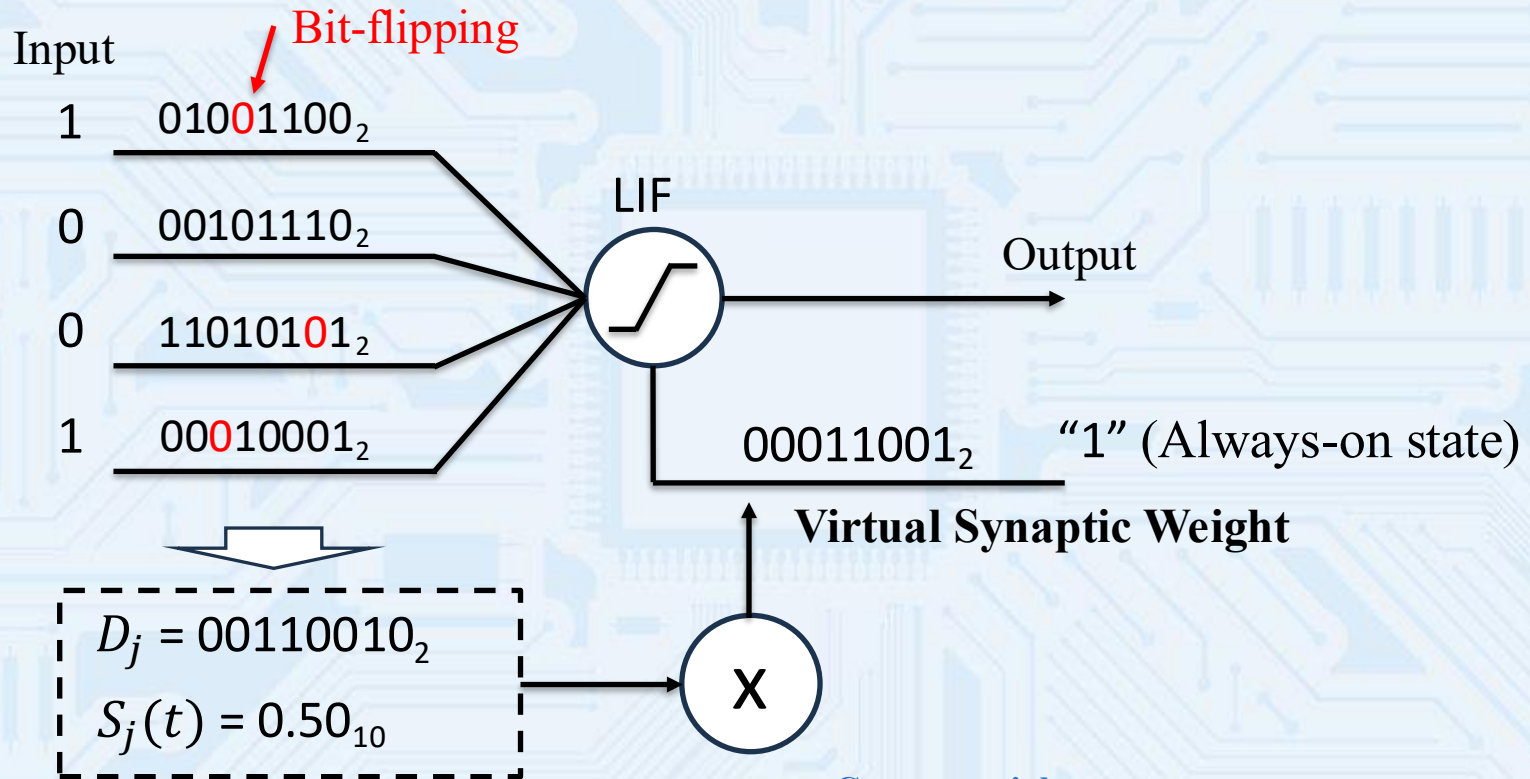
Proposed Methodology

- Illustration of how SEAL works



Proposed Methodology

- Illustration of how SEAL works



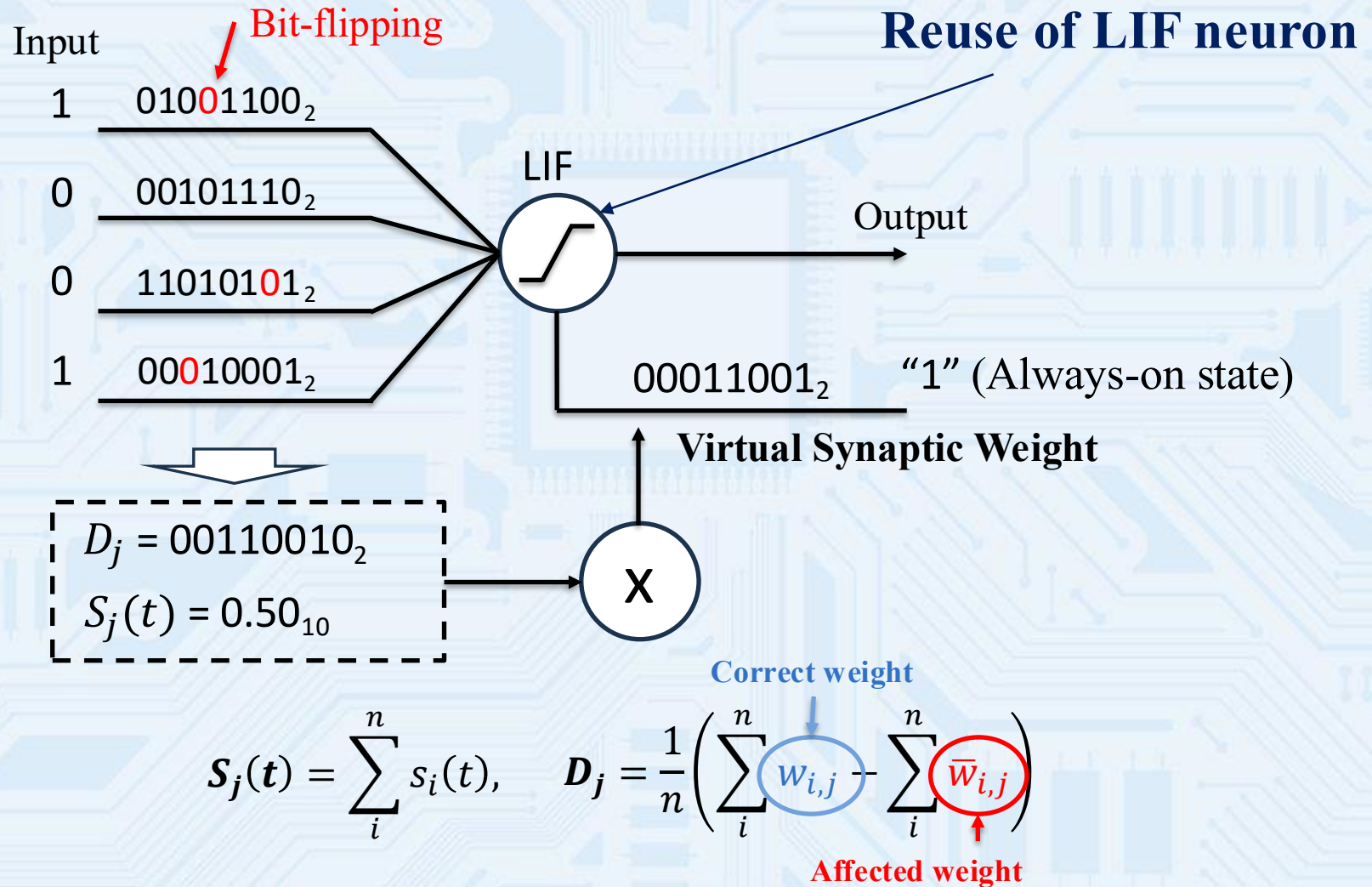
Correct weight

Affected weight



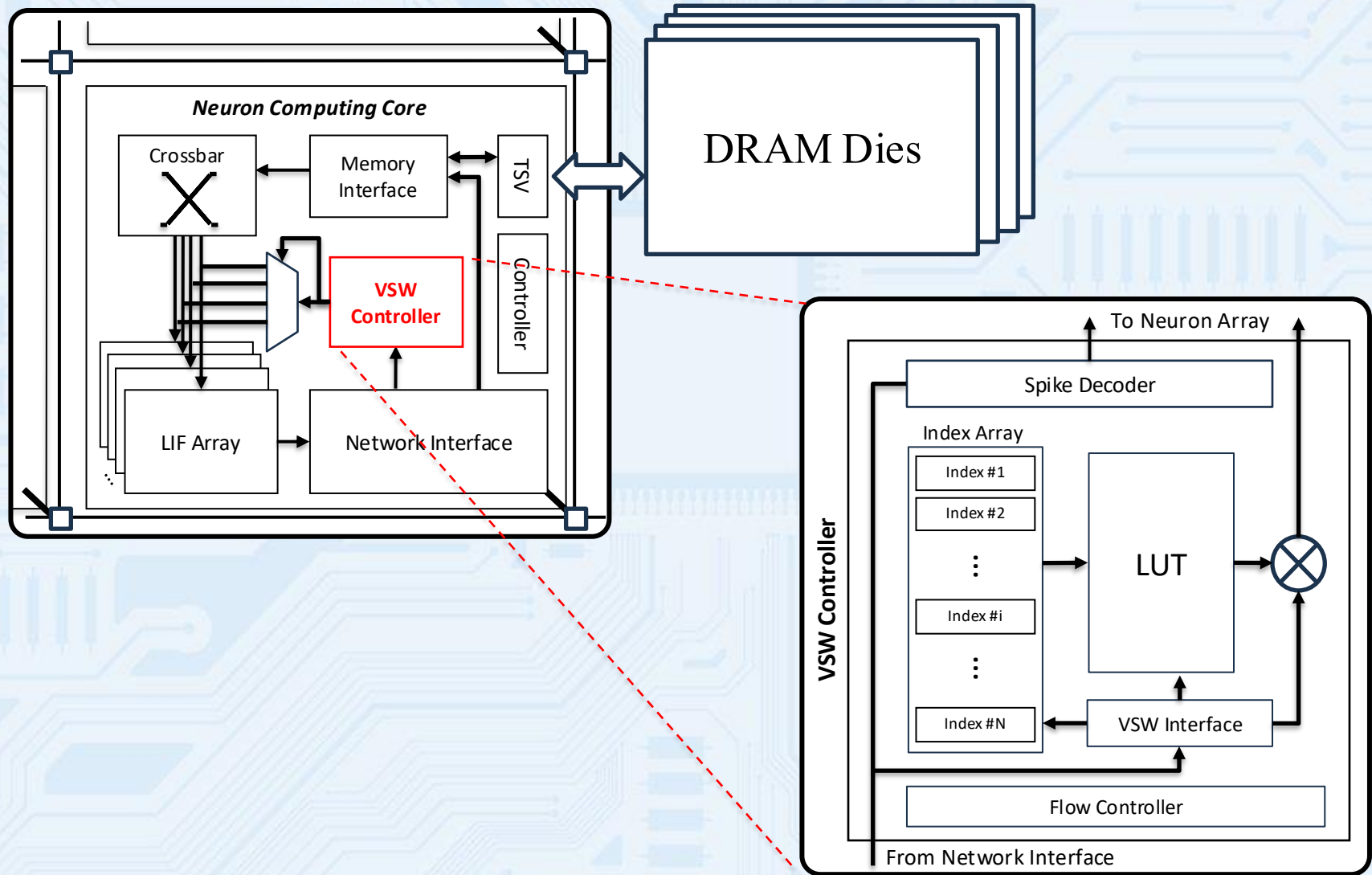
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- Illustration of how SEAL works



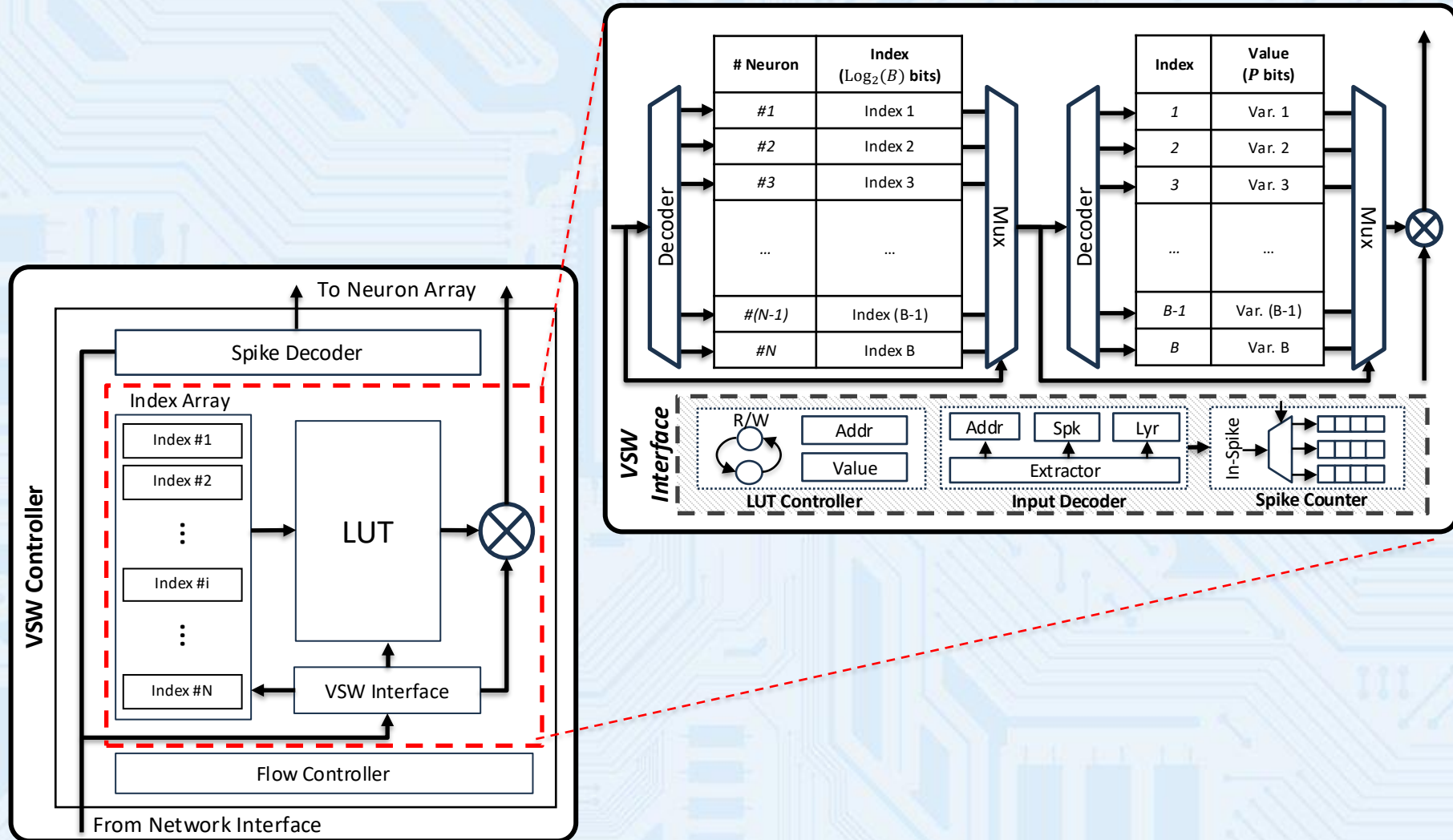
Proposed Methodology

- Overview of hardware implementation



Proposed Methodology

- Overview of hardware implementation



Proposed Methodology

- LUT implementation strategy
 - Use binning approximation

Exact D_j (16 bits)

Neurons $\left[\begin{array}{c} 5.022 \\ 4.817 \\ 4.251 \\ 7.338 \\ \vdots \\ \vdots \\ 4.990 \\ 6.174 \\ 3.907 \end{array} \right]$

Bin Assignment Table

ID	Range	Base Val.
0	3.51 – 4.00	3.51
1	4.01 – 4.50	4.01
2	4.51 – 5.00	4.51
3	5.01 – 5.50	5.01
4	5.51 – 6.00	5.51
5	6.01 – 6.50	6.01
6	6.51 – 7.00	6.51
7	7.01 – 7.50	7.01

Index (3 bits)

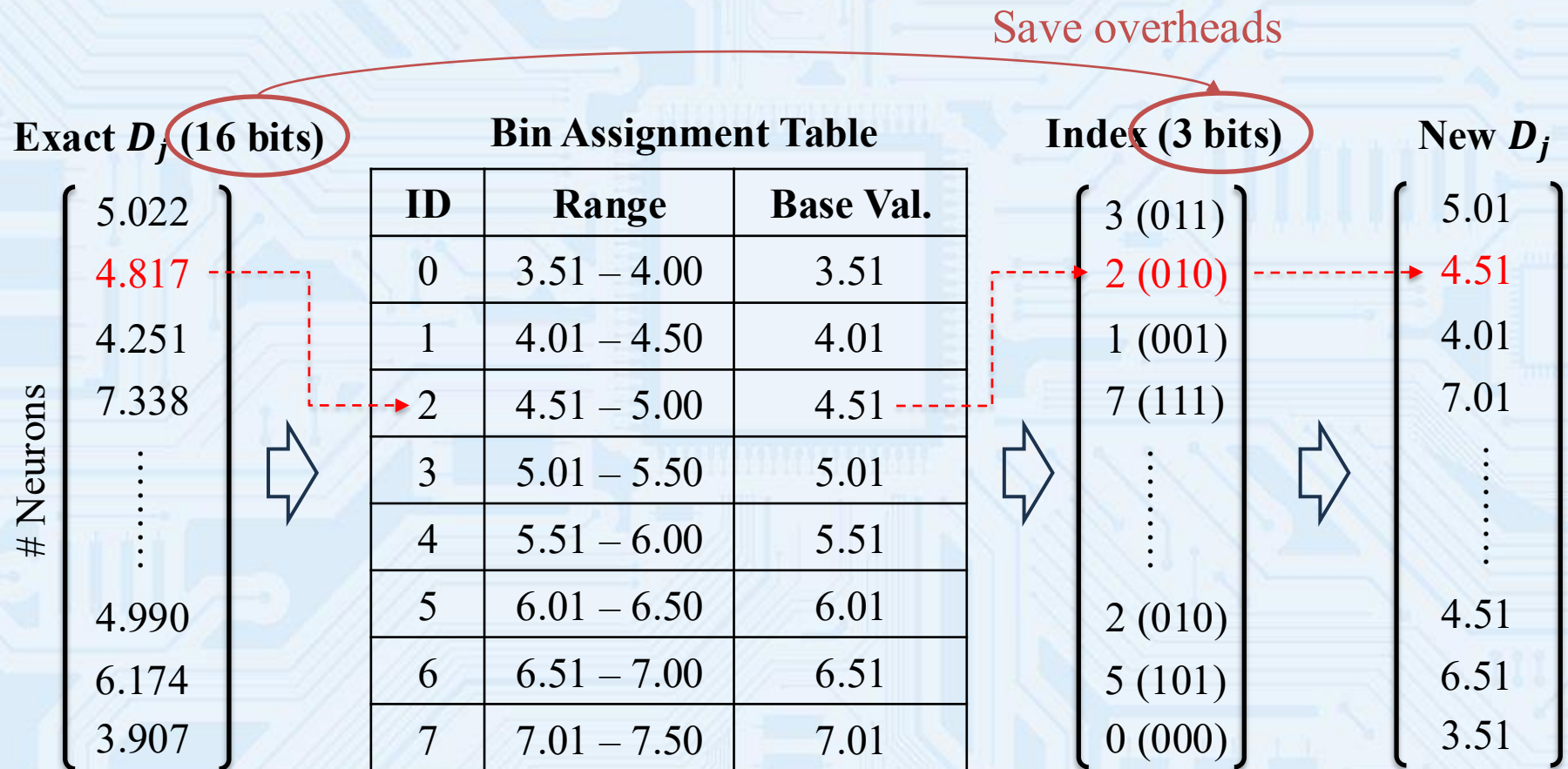
$\left[\begin{array}{c} 3 \text{ (011)} \\ 2 \text{ (010)} \\ 1 \text{ (001)} \\ 7 \text{ (111)} \\ \vdots \\ \vdots \\ 2 \text{ (010)} \\ 5 \text{ (101)} \\ 0 \text{ (000)} \end{array} \right]$

New D_j

$\left[\begin{array}{c} 5.01 \\ 4.51 \\ 4.01 \\ 7.01 \\ \vdots \\ \vdots \\ 4.51 \\ 6.51 \\ 3.51 \end{array} \right]$

Proposed Methodology

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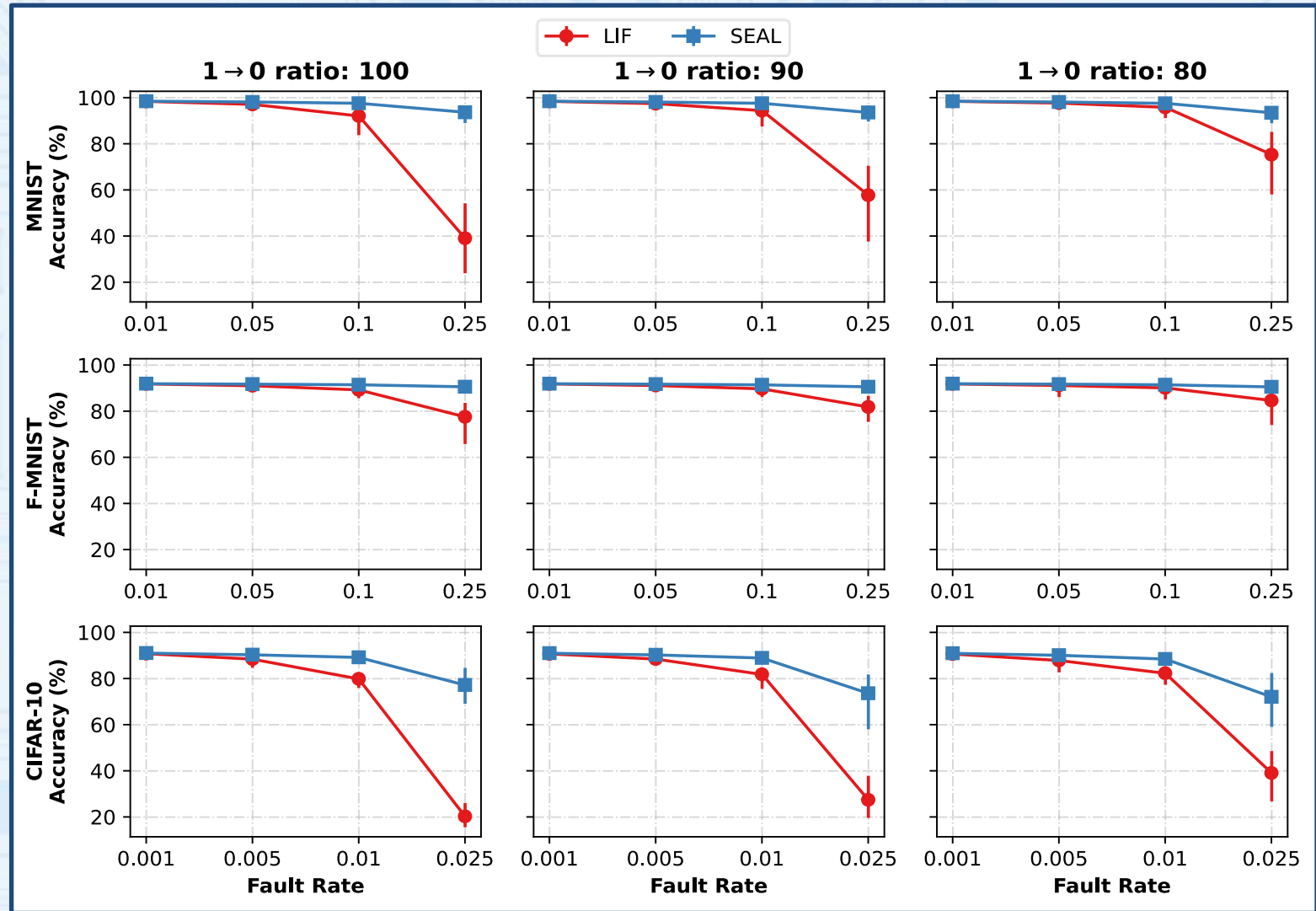
Evaluation

- Accuracy assessment:
 - Network architecture and benchmark
 - *MNIST* (MLP), *F-MNIST* (VGG5), *CIFAR-10* (VGG16)
 - Insert single-bit faults randomly to trained weights
 - **1-to-0** bit-flipping @ 100%, 90%, 80% ratio
 - Change granularity of binning approximation
- Hardware evaluation:
 - Logic synthesis with Synopsys Design Compiler
 - 45nm technology node
 - Memory modeling:
 - DESTINY[6] tool to model 3D-DRAM with 45nm node



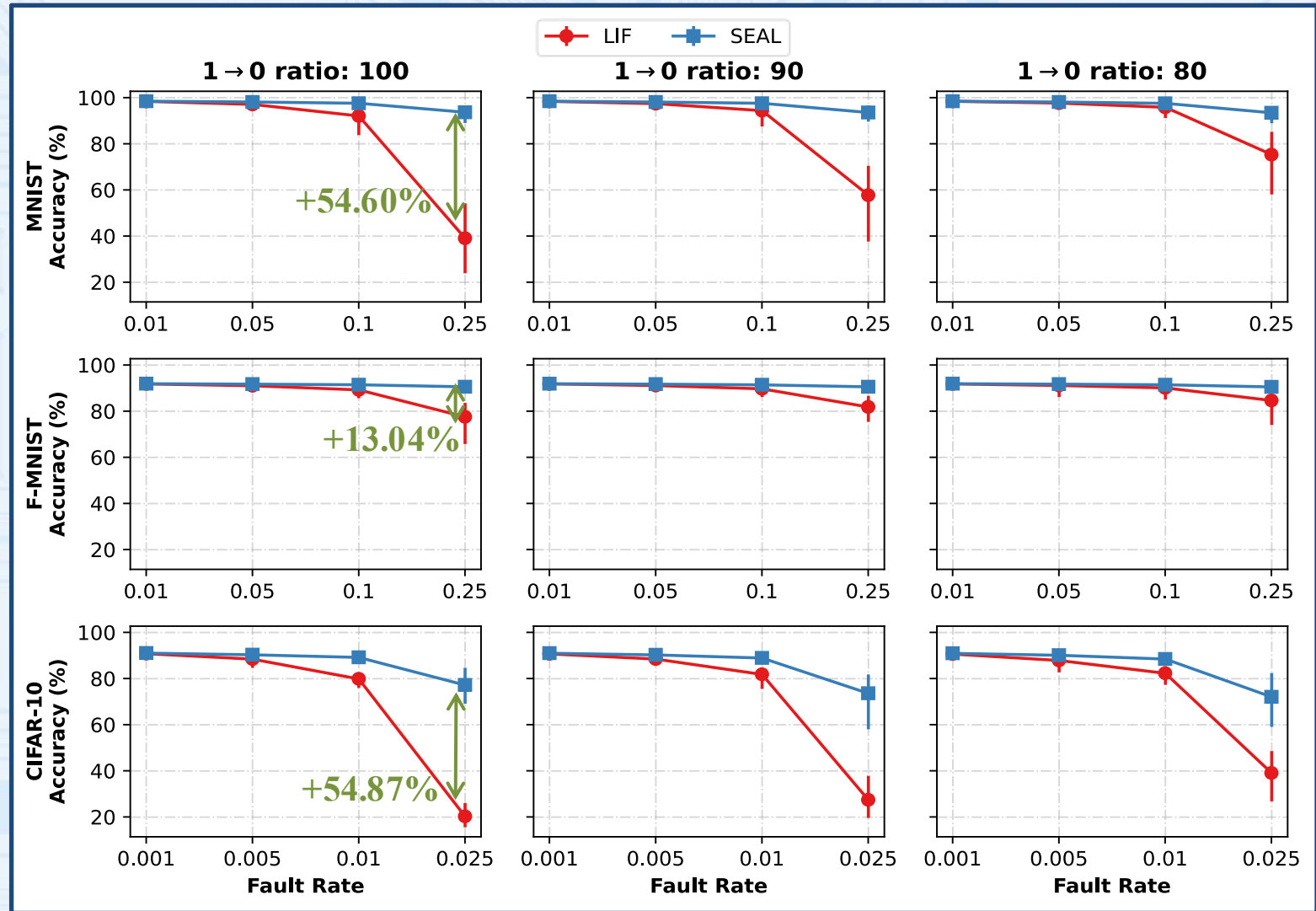
Evaluation

- Impact on benchmark accuracy



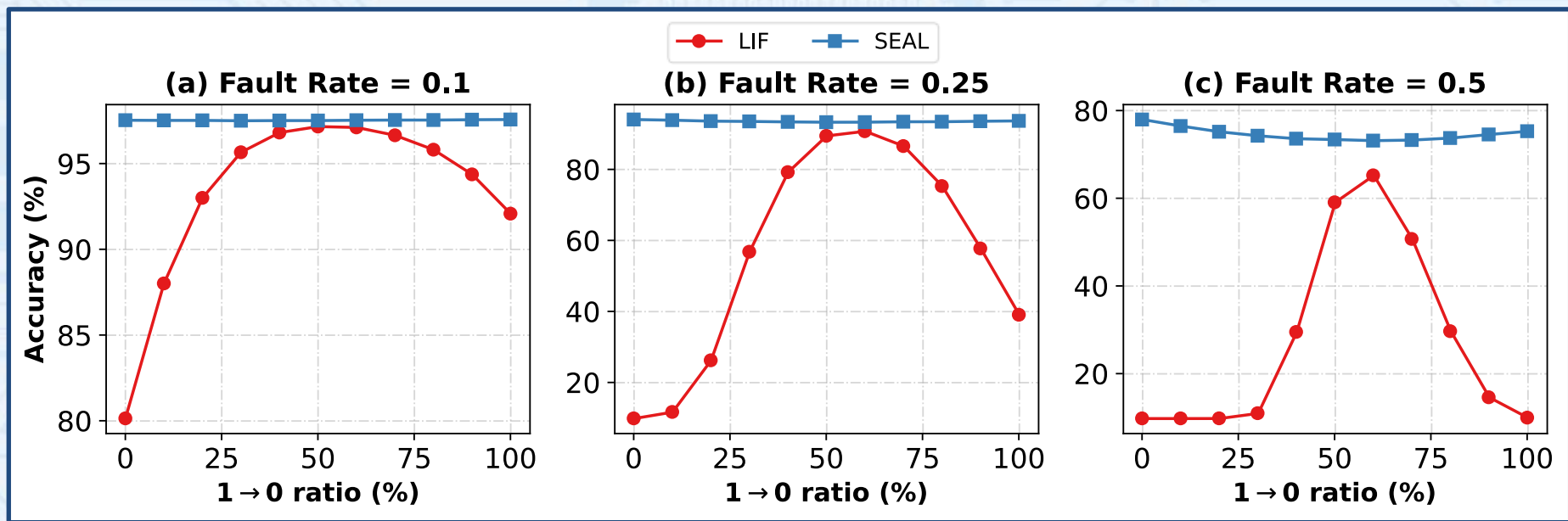
Evaluation

- Impact on benchmark accuracy



Evaluation

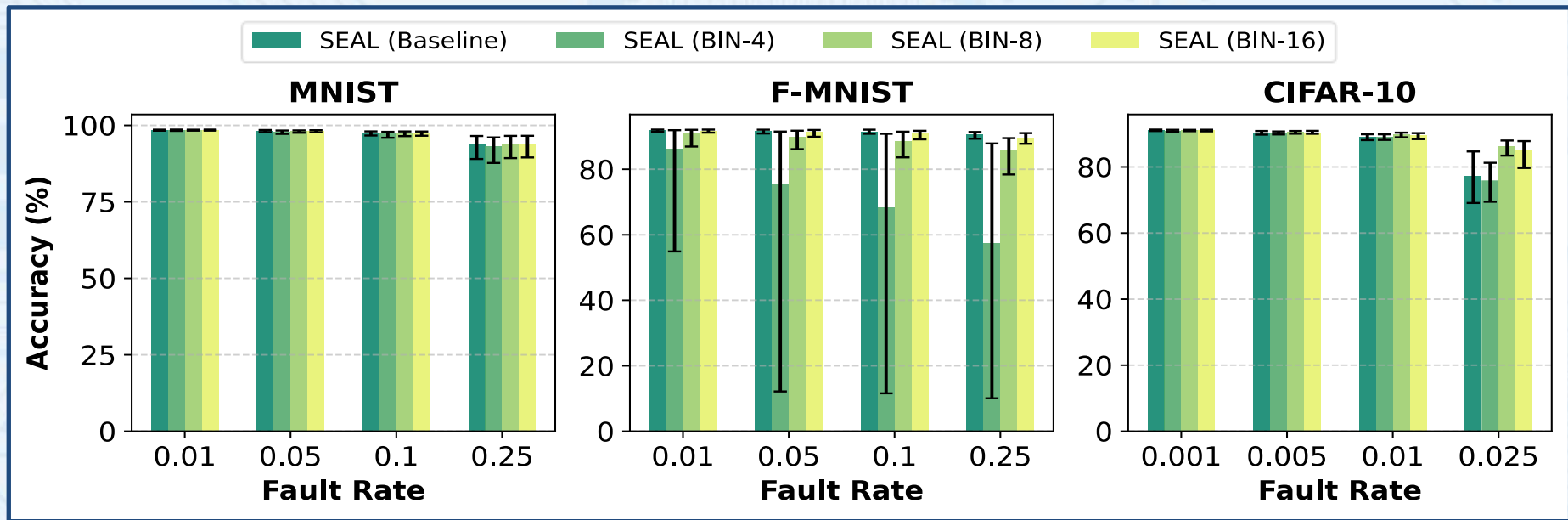
- Characteristics of accuracy
 - SEAL can be both excitatory and inhibitory factors



=> Constantly providing better accuracy than the LIF neuron

Evaluation

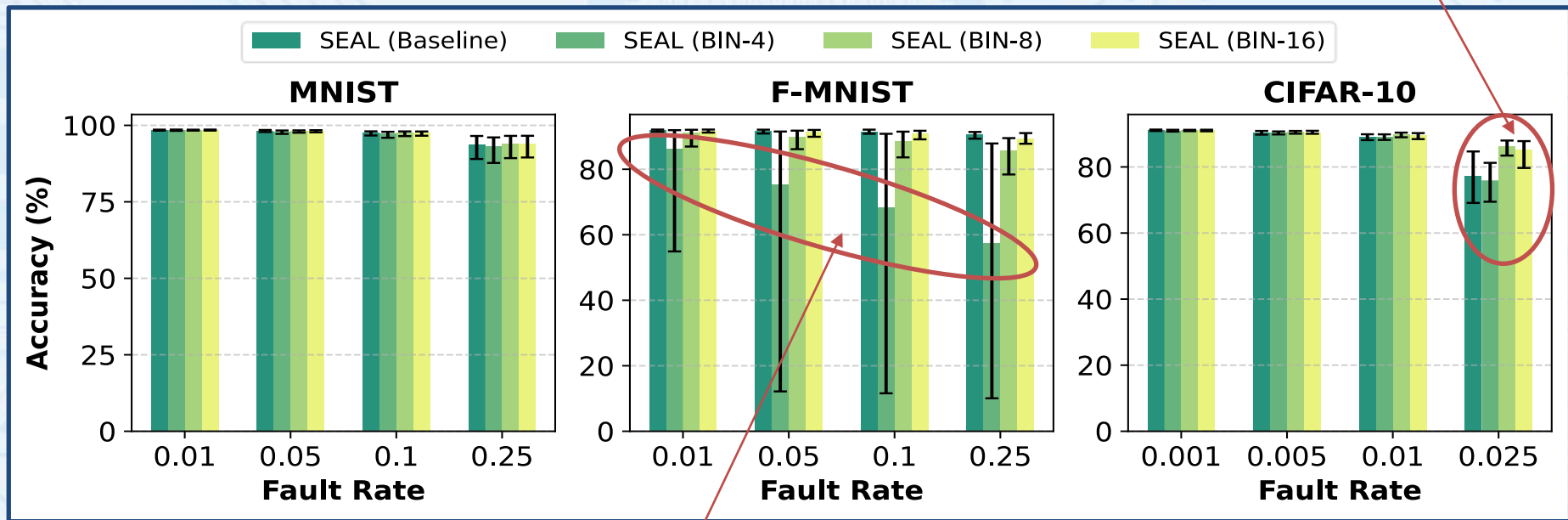
- Impact of binning approximation
 - Depends on network architectures & workloads



Evaluation

- Impact of binning approximation
 - Depends on network architectures & workloads

Even improve accuracy



Significant drop



Evaluation

- Hardware evaluation
 - Modest overheads

TABLE I
HARDWARE OVERHEAD COMPARISON FOR DIFFERENT MODELS

Configuration	$784 \times 256 \times 256 \times 10$		$784 \times 1024 \times 1024 \times 10$	
	Area	Energy	Area	Energy
BIN-4	1.41%	3.55%	1.47%	3.41%
BIN-8	1.61%	5.16%	1.73%	4.79%
BIN-16	1.88%	6.72%	2.01%	5.59%



Evaluation

- Comparison to prior arts

TABLE II
PERFORMANCE COMPARISON BETWEEN THE PROPOSED METHODOLOGY
AND PRIOR WORKS

	ICCAD'21 [1]	DAC'22 [2]	TCPMT'24 [3]	Our work
Network	784:400	784:400	784:256:256:10	784:256:256:10
Dataset	MNIST	MNIST	MNIST	MNIST
Methodology	Fault-Aware Mapping	Bound-and-Protect	Swapping Weights	Virtual Synaptic Weights
On-the-fly	No	Yes	No	Yes
Baseline Acc.	$\sim 86\%^1$	$\sim 86\%^1$	97.78%	98.54%
Acc. Loss (@FR=0.1)	$\sim 10\%^1$	$\sim 12\%^1$	0.01–0.24% ²	0.96–1.24% (0.00–0.09%) ²

¹ Accuracy loss is calculated based on the provided images.

² Without errors in MSBs.



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Conclusion & Future Works

- SEAL neuron compensates for retention failure
- Virtual Synaptic Weight (VSW) minimizes the modification from the LIF neuron
 - VSW controller is independent of neuron types
- Results show enhanced fault-tolerance and reasonable HW overhead
- Future works include thermal-aware fault-tolerance based on SEAL



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- [1] A. Gholami *et al.*, “AI and Memory Wall,” IEEE Micro, 2024.
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- [3] R. V. W. Putra *et al.*, “ReSpawn: Energy-Efficient Fault-Tolerance for Spiking Neural Networks considering Unreliable Memories,” in 2021 IEEE/ACM International Conference on Computer-Aided Design (ICCAD). IEEE, 2021, pp. 1–9.
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- [5] N.-D. Nguyen *et al.*, “NOMA: A Novel Reliability Improvement Methodology for 3-D IC-based Neuromorphic Systems,” IEEE Transactions on Components, Packaging and Manufacturing Technology, 2024.
- [6] M. Poremba *et al.*, “DESTINY: A tool for modeling emerging 3D NVM and eDRAM caches,” in 2015 Design, Automation & Test in Europe Conference & Exhibition (DATE). IEEE, 2015, pp. 1543–1546.



THANK YOU

